Solar Installation Report

Project	Shrestha house, Biratnagar
Client	-
Location	Biratnagar
Prepared by	-
Report date	2026-10-30
Software	ElectricalCalc AI v0.4.0
Review status	DRAFT - not reviewed
Reviewer	Not reviewed

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Summary in plain words

- You need 4 solar panels of 550 W (2.20 kW in total), needing about 10 m² of roof.
- They should make about 3,348 units (kWh) a year - about 279 a month on average; you use about 199 units a month.
- Inverter (turns solar power into normal 230 V power): about 1.8 kW.
- Battery: 57 Ah at 48 V (about 2.7 kWh), lasting about 12 years.
- Estimated cost: about Rs 271,807 (indicative Nepal prices - get installer quotes).
- Estimated saving on your NEA bill: about Rs 21,027 per year.
- The system pays for itself in about 18.2 years.
- This is a first estimate. A qualified electrical engineer must check it, the site and the equipment before anything is bought or installed.

Scope and notes

Roof: Flat roof - panels go on a metal frame. Panels modelled at 26° tilt, azimuth 0° (0 = south). PVGIS v5.3 (PVGIS-ERA5, 2005-2023) at 26.4525, 87.2718; tilt 26°, azimuth 0° (0 = south); 14.0% system loss.

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A. What to buy (shopping list)

Buy equipment that meets NEPQA-2025 (AEPC) and ask the seller for the datasheets. Ratings below are the minimum - equal or higher is fine unless stated.

Item	Specification	Qty	Why / rule
Solar PV module	550 Wp mono/TOPCon, Voc 49.5 V, Vmp 41.6 V, Isc 14 A, efficiency ≥ 21 %, power warranty ≥ 80 % at 25 yr, IEC 61215/61730	4	Array 4 in series × 1 string(s) = 2.20 kWp; NEPQA-2025 module floors
Mounting structure	Hot-dip galvanised / aluminium tilt frame with anchor bolts, tilt as designed	1 row(s) of up to 4	Boxwell, Solar Electricity Handbook §123; roof must carry the weight - check structure
Mounting rails	Aluminium PV rail (2 per row of panels)	10 m	2 rails under each panel
End clamps / mid clamps	Matching panel frame thickness	4 end / 6 mid	4 end clamps per row, 2 mid clamps between neighbouring panels
Roof feet / anchors	Stainless fixings, sealed against leaks	10	One fixing about every 1.2 m of rail
PV cable (red and black)	4 mm² solar cable EN 50618 (H1Z2Z2-K), UV resistant, 1500 V DC	18 m red + same black	Cable schedule; book §87, §110
MC4 connector pairs	Same brand as panel connectors (never mix brands)	4	Field-made string ends + spares
DC isolator (PV)	600 V DC, ≥ 25 A, 2-pole, lockable	2	Rated above cold string Voc 211 V and 1.25 × Isc; book §79, §80, §109
DC surge protection device (Type 2, PV)	Ucpv ≥ 600 V DC, In ≥ 20 kA	1	Lightning-induced surges are common in Nepal; above cold string Voc
Hybrid inverter (low-voltage battery)	2 kW continuous, surge ≥ 0.8 kW, battery 48 V, PV max 500 V, MPPT 125-425 V, pure sine 230 V 50 Hz, grid + backup outputs	1	Carries backup loads (×4/3, book §91) and the whole array
Battery	LiFePO4 48 V 100 Ah with BMS, ≥ 3000 cycles @ 80 % DoD, round-trip ≥ 85 %, warranty ≥ 5 yr, communication with inverter (CAN/RS485)	1	1 in series × 1 in parallel = 4.8 kWh; needs 57 Ah at 48 V (book §89)
Battery cable (red and black)	25 mm² flexible copper, crimped lugs	5 m each colour	Cable schedule; keep as short as possible (§125)
Battery fuse (DC, high breaking capacity)	63 A NH/ANL/class-T DC fuse	1	≥ 1.25 × max battery current 46 A; on the + terminal so all current passes (§109)
Battery isolator / DC breaker	63 A, 48 V DC rated	1	Lets you disconnect the battery safely (book §79)
AC MCB (inverter output)	16 A, C-curve, 1-pole + N	1	≥ 1.25 × inverter AC current 8.7 A; cable sized to carry it
RCD / RCBO	30 mA, Type A (Type B if the inverter manual requires)	1	Shock protection on AC circuits (§109)

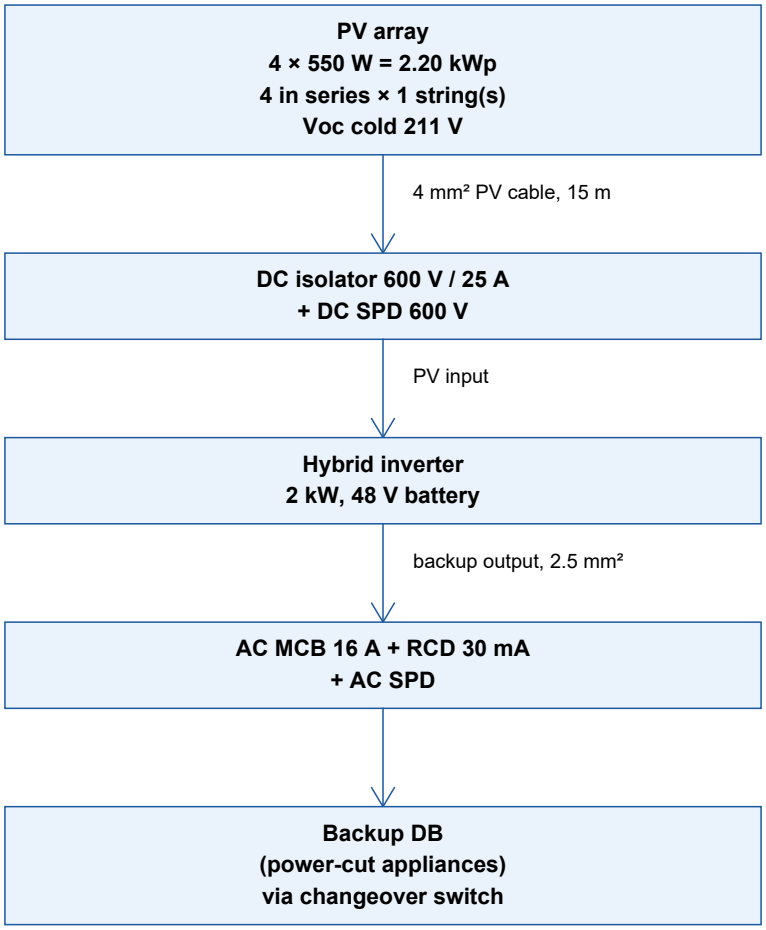
Item	Specification	Qty	Why / rule
AC surge protection device (Type 2)	275 V AC, In ≥ 20 kA	1	Protects inverter and loads
AC cable	2.5 mm² copper (phase + neutral) + 2.5 mm² earth	10 m	Cable schedule
Backup distribution board	Separate small DB for the appliances needed in power cuts	1	Hybrid inverter backup output feeds only these circuits (book §29-§30, §160)
Manual bypass / changeover switch	16 A, 2-pole, 3-position (inverter / off / grid)	1	Keeps the backup circuits powered from NEA if the inverter is serviced
Earth electrode	Copper-bonded rod ≥ 1.5 m (or copper plate earth pit per local practice), inspection pit	1	Earth the array frame, inverter and battery negative (§109, §121, §124)
Earth cable (green/yellow)	6 mm² frame bonding, 16 mm² to electrode, 2.5 mm² PE	23 m total	Cable schedule
Conduit / cable tray, glands, ties	UV-resistant conduit outdoors; glands sized to cables	as required	Protect cable runs (§110)
Warning labels	'Solar DC - live in daylight', isolator labels, dual-supply label at DB and meter	1	IEC 60364-7-712 practice
NEA bidirectional (import/export) meter	Supplied/approved by NEA after net-billing application	1	Needed to be credited for exported units; apply at your NEA distribution centre

B. How the panels and batteries are connected

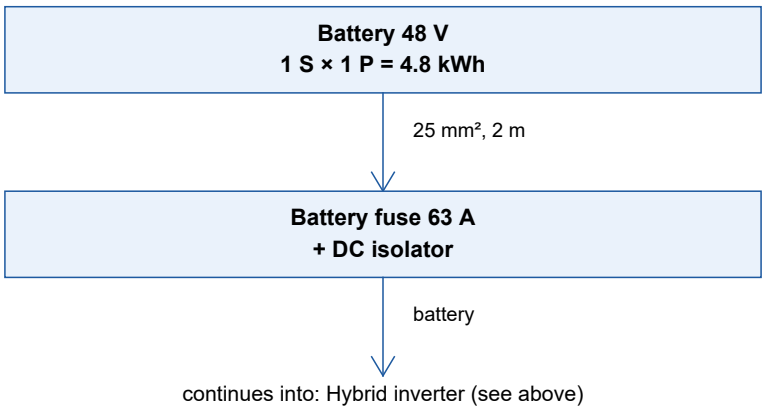
- Modules per string: 4
- Strings: 1
- Panels: 4
- Kwp: 2.2
- String voc stc v: 198.0
- String voc cold v: 210.86999999999998
- String vmp stc v: 166.4
- String vmp hot v: 145.912
- N min: 4
- N max: 9
- Pv input: hybrid inverter
- Pv input limits: max 500 V, MPPT 125-425 V
- T min c: -1.0
- T max c: 43.0
- String fuses: None
- Battery series: 1
- Battery parallel: 1
- Battery kwh: 4.8

C. Wiring diagram

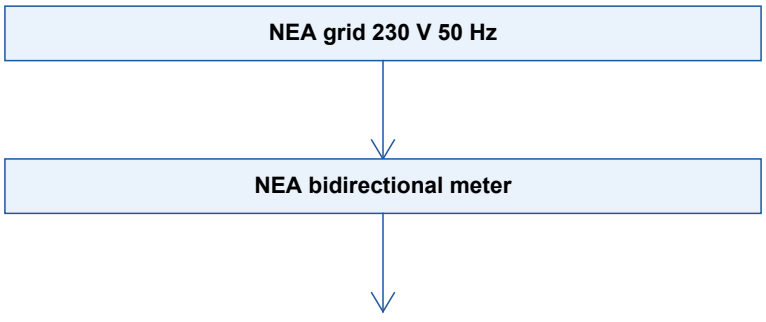
Path 1

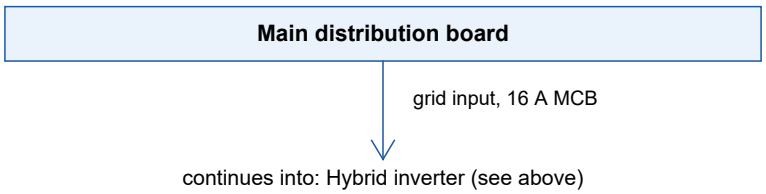


Path 2

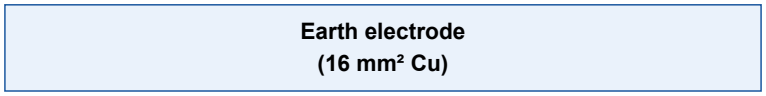


Path 3





Path 4



Every connection (wire from this table)

From	To	Cable / protection
PV array	DC isolator 600 V / 25 A	4 mm² PV cable, 15 m
PV array	Earth electrode	6 mm² frame bonding
Battery 48 V	Battery fuse 63 A	25 mm², 2 m
Battery 48 V	Earth electrode	battery − earthed (§109)
DC isolator 600 V / 25 A	Hybrid inverter	PV input
Battery fuse 63 A	Hybrid inverter	battery
Hybrid inverter	AC MCB 16 A + RCD 30 mA	backup output, 2.5 mm²
AC MCB 16 A + RCD 30 mA	Backup DB	
NEA grid 230 V 50 Hz	NEA bidirectional meter	
NEA bidirectional meter	Main distribution board	
Main distribution board	Hybrid inverter	grid input, 16 A MCB
Hybrid inverter	Earth electrode	2.5 mm² PE

Each arrow is a cable: the label gives its size and the fuse/breaker that protects it.

D. Cables, fuses and breakers

Circuit	From → To	Length (m)	Design current (A)	Protective device (A)	Size for current (mm²)	Size for voltage drop (mm²)	Chosen (mm²)	Voltage drop (%)
PV string cable (each string, + and −)	Array → hybrid inverter	15.0	17.5	17.5	2.5	4	4	1.31
Battery cable (+ and −)	Battery bank → fuse/isolator → inverter	2.0	45.8	63	25	10	25	0.33
Inverter AC output	Inverter → distribution board	8.0	8.7	16	2.5	1.0	2.5	0.54
PV frame bonding (earth)	Each panel frame / rail → earth bar	-	-	-	-	-	6	-
Main earthing conductor	Main earth bar → earth electrode	10.0	-	-	-	-	16	-
Inverter / DB protective earth (PE)	Inverter → DB earth bar	8.0	-	-	-	-	2.5	-

Chosen size = the larger of the current-rating size and the voltage-drop size.

E. Safety first

- Panels produce dangerous DC voltage whenever there is daylight - there is no off switch on a panel (§119, §132).
- Never unplug MC4 connectors or open a DC fuse holder while current flows - switch off the isolator first.
- A battery short circuit releases huge current: fire, burns, explosion. Fuse it, insulate tools, remove jewellery.
- Work at height only with a ladder footed by a helper, a harness on steep roofs, and never in rain or wind.
- Lead-acid batteries contain acid and release hydrogen - ventilate, no flames, wear goggles and gloves.
- Connection to the building wiring and the NEA grid must be done/checked by a licensed electrician and approved by NEA; never back-feed the grid without approval (danger to line workers).
- Use NEPQA-2025 compliant equipment (AEPC) so warranties and any subsidy remain valid.

F. Step-by-step installation

1. Before you start - safety and tools (§118-§120)

- 1. Read every manual (inverter/controller, battery) first - the settings matter as much as the wiring.
- 2. Tools: digital multimeter (DC up to 1000 V), DC clamp meter, insulated screwdrivers, crimping tool for MC4 and battery lugs, torque wrench, drill, spirit level, compass (phone app), ladder or scaffold.
- 3. Protection: insulated gloves (Class 0 for arrays above 500 V, Class 00 below), safety glasses, safety shoes, harness for roof work. Keep a first-aid kit; for lead-acid add eye wash and acid spill kit.
- 4. Never work alone on a roof. Keep children and animals away and tell people nearby you are working with high voltage.
- 5. Remove rings, watches and bracelets before touching batteries - a short circuit can weld metal.
- 6. This array reaches about 211 V DC - lethal. Wear insulated gloves when handling PV connectors (§126).

2. Site check (§63-§73, §121)

- 1. Confirm the roof spot is free of shade from about 9 am to 3 pm all year (trees, water tanks, neighbouring floors - Nepali rooftops grow!).
- 2. Check the roof can carry the panels and frame; on tin roofs fix into the purlins, never only the sheet.
- 3. Choose the inverter/battery place: dry, shaded, ventilated, close to the battery (within ~1-2 m, §125) and near the distribution board. Batteries off a cold concrete floor on a shelf (§124).
- 4. Mark the cable route and the earth-pit location.

3. Mount the structure and panels (§123)

- 1. Fix the tilt frame facing due south at the designed tilt; tighten all bolts.
- 2. Unpack each panel and look for cracked glass; measure each panel's open-circuit voltage in sun - expect about 49.5 V (cooler days higher).
- 3. Clamp the panels to the rails (end clamps at row ends, mid clamps between panels).
- 4. Keep panels covered with an opaque sheet, or leave the strings unconnected, while you wire - panels make electricity whenever light falls on them.

4. Wire the PV strings and check polarity (§106, §123)

- 1. Connect 4 panels in series per string (+ of one to − of the next) using the panel leads.
- 2. Make 1 string(s). Use only matching MC4 connectors, crimped with the proper tool.
- 3. Measure each string's open-circuit voltage before going further: expect about 198 V at 25 °C (up to 211 V on a cold morning). All strings should read within ~2 % of each other.
- 4. Check polarity with the meter (red + / black −). A reversed string can destroy the inverter.
- 5. Run the PV cables (4 mm² red and black) in UV-resistant conduit to the DC isolator; keep + and − close together to limit loops (lightning).
- 6. Leave the DC isolator OFF.

5. Earthing (§109, §121, §124)

- 1. Make the earth pit/rod and connect a 16 mm² green/yellow cable to the main earth bar.
- 2. Bond every panel frame and rail with 6 mm² green/yellow cable (or approved bonding clips).
- 3. Connect the inverter and distribution-board earth terminals to the earth bar.
- 4. For battery systems, earth the battery negative as the inverter manual allows (many LiFePO4 hybrid inverters require it to stay floating - follow the manual).
- 5. Measure and record the earth resistance; improve the pit (deeper rod, second rod) if it is high.

6. Install the battery (§124)

- 1. Place the battery upright on a sturdy shelf, ventilated, away from sparks, sun and children; label it with the installation date.
- 2. Connect 1 module(s) in series and 1 string(s) in parallel with the maker's interconnect cables; use equal cable lengths for parallel strings.
- 3. Tighten every terminal to the torque on the datasheet; protect terminals from short circuits.
- 4. Fit the 63 A battery fuse on the + cable close to the battery, then the battery isolator - leave it OFF.
- 5. Measure the bank voltage: LiFePO4 48 V-class banks rest at about 51-53 V; lead-acid about 51-53 V when charged.

7. Mount the inverter / controller (§125)

- 1. Mount vertically on a non-flammable wall with the clearances in its manual (it gets hot).
- 2. Connect the battery cables (25 mm²) to the inverter: negative first, check polarity twice, then positive.
- 3. Connect the PV DC isolator output to the PV input terminals - still with the isolator OFF.
- 4. Set the battery type and voltages in the inverter/controller menu before switching on (§127).

8. AC connection - STOP: licensed electrician (§119, §126)

- 1. The inverter AC output goes through a 16 A MCB, a 30 mA RCD and an AC surge protector to the backup distribution board via the changeover switch.
- 2. Connecting to the house wiring must be done or checked by a licensed electrician.
- 3. For net billing: apply at your NEA distribution centre BEFORE connecting to the grid; NEA approves the inverter (anti-islanding) and fits a bidirectional meter. Exports are not credited before approval.

9. Switch on (§125, §127-§128)

- 1. Turn ON the battery isolator - the inverter/controller should wake up and show battery voltage.
- 2. Turn ON the PV DC isolator in daylight - check the PV voltage and charging/feeding power on the display.
- 3. Turn ON the AC MCB and test with a small load such as a lamp.
- 4. Shutdown order is the reverse: AC MCB off → PV isolator off → battery isolator off. Never open DC connectors under load.

10. Commissioning tests and first week (§127-§128, IEC 62446)

- 1. Fill in the commissioning record table in this report (expected values are given).
- 2. Insulation test of each string to earth at 500 V: minimum 1 MΩ.
- 3. Simulate a power cut (switch off the NEA main) and check the backup circuits keep running.
- 4. For battery systems, let the battery charge fully for a day before heavy use (§128).
- 5. Watch the daily kWh for a week; compare with the monthly estimate in this report.

G. Testing before switching on (write the measured values)

Test	Expected	Measured	OK?
String 1 open-circuit voltage (Voc)	≈ 198 V at 25 °C (max 211 V cold)		
String 1 short-circuit current (clamp meter, sunny)	≤ 14 A (≈ Isc × sunlight/1000 W/m²)		
Polarity of every string and the battery	Correct (+ to +)		
Insulation resistance, string + and − to earth	≥ 1 MΩ		
Earth electrode resistance	As low as practical - record value		
Battery bank voltage at rest	48-55 V		
Inverter AC output voltage / frequency	230 V ±10 %, 50 Hz (NEPQA)		
RCD trip test button	Trips		

H. Looking after the system

When	Task
Weekly	Glance at the inverter/controller display: daily kWh, battery state, any fault codes; keep a simple logbook (§139).
Monthly (dry season: Oct-May)	Clean panels early morning with soft brush and plain water - Nepal dust and brick-kiln soot cut output; no detergent abrasives.
Monthly	Check battery area: ventilation, no leaks, dust off battery tops; LiFePO4: check BMS alarms. Lead-acid: water level (§139).
Before monsoon	Check roof fixings, conduit seals and junction boxes for water ingress; clear drains around the earth pit.
Every 6 months	Re-torque battery and DC terminals (power off); look for hot or discoloured connectors; test the RCD button.
Yearly	Measure string voltages and compare with the commissioning record; test earth resistance; check SPD status windows; balance lead-acid batteries (§139).
After lightning / storm	Check SPD indicators and replace tripped cartridges; inspect panels for damage.

I. If something goes wrong

Symptom	Likely cause	What to check
Battery runs flat / power cuts off early	Using more energy than designed (§134)	Logbook or inverter daily use vs the design figure; reduce load or add panels/battery.
Lower solar output than expected	Dirt, new shade, wrong angle (§135)	Clean panels; look for new shade (tree, tank, new floor next door); compare string voltages.
Strange or intermittent faults, worse in heat/cold	Loose or damaged connection (§136)	With power off: tighten terminals, inspect MC4s for burn marks, measure voltage drop along cables.
Power fails when a big appliance starts	Inverter surge too small or weak battery (§137-§138)	Compare motor starting current with inverter surge rating; check battery voltage under load.
One battery hotter or lower voltage than others	Failing cell (§137)	Voltage difference > 0.7 V needs balancing; ≥ 2 V means a failed battery - replace it.
Inverter trips on hot afternoons	Overheating / poor ventilation (§138)	Clear space around the inverter, keep it out of direct sun, check its fan.
Inverter trips in rain	Earth leakage / insulation fault (§138)	Insulation-test strings; look for water in junction boxes and damaged cable.
Grid-tied inverter shows 'grid over-voltage' at midday	High NEA voltage plus cable rise	Measure grid voltage; a larger AC cable reduces voltage rise; ask NEA about transformer tap.

References: Boxwell, Solar Electricity Handbook §104-§112 (design, protection), §117-§130 (installation), §131-§139 (troubleshooting, maintenance); IEC 62548-1 (PV array design), IEC 62446-1 (testing), IEC 60364-5-52 (cable current ratings), IEC 60364-7-712 (PV installations); NEPQA-2025 and Amendment-1 (AEPC); Copper/PVC in conduit, IEC 60364-5-52 method B1, 2 loaded conductors (conservative)

1. Home solar - hybrid with power-cut backup

Input parameters

System type	hybrid backup
Charge controller / MPPT	MPPT
Panel rating	550.00 W
Inverter efficiency	94 %
Temperature loss	10.87 %
Shading loss	0 %
Battery type	LiFePO4 (lithium)
Depth of discharge	90 %

Calculation method and equations

- Daily Wh = Σ (W × qty × hours) (§48)
- Design Wh = DC Wh + AC Wh ÷ η_inverter (§50)
- Array W = Design Wh ÷ sun hours (annual average) ÷ (η_controller × (1 − temperature loss) × (1 − shading)) (§56-§59)
- Panels = ceil(Array W ÷ panel Wp)
- Battery Wh = backup appliances W (÷ η_inverter for AC) × power-cut hours ÷ η_battery [modern/derived]
- Battery Ah = Battery Wh ÷ system V ÷ depth of discharge (§89)
- Current = W ÷ system V (§86); Cable mm² = L × I × 0.04 ÷ (V ÷ 20) (§87)
- System V: lowest of 12/24/48 V keeping that cable ≤ 6 mm² (§85); 12 V only up to ~3 kW (§91)
- Controller A = array W ÷ system V × 1.25 [modern margin - the book gives none, §90]
- Inverter W ≥ max(4/3 × backup appliances (§91), installed array ÷ 1.2) [modern]

Results (calculated by deterministic code)

Appliance energy per day	6,550.00 Wh/day
Energy the solar must supply per day (after losses)	6,968.09 Wh/day
Everything switched on at once	1,760.00 W
Sun hours used (annual average)	5.198 kWh/m²/day
Solar array needed	1,671.27 W
Panels of 550 W	4 panels
Installed solar array	2,200.00 W
Roof area for panels	10.32 m²
Battery / system voltage	48 V
Energy the battery must deliver	2,463.61 Wh
Battery capacity at 48 V	57.03 Ah
Battery nominal capacity	2.737 kWh
Charge controller rating (array current × 1.25)	57.29 A
Battery/array cable by the book's formula (5 % drop)	2.292 mm²
Estimated battery life (one cycle per day, capped)	12 years
Inverter continuous rating	1,833.33 W
Estimated solar generation per year	3,347.97 kWh/yr
Appliance use per year	2,390.75 kWh/yr

Appliance list (power analysis)

Appliance	W	Qty	Hours/day	Wh/day	Supply	Backup	Motor
Refrigerator (typical - check label)	150.0	1.0	10.0	1500.0	AC 230 V	Yes	Yes

Appliance	W	Qty	Hours/day	Wh/day	Supply	Backup	Motor
LED bulb (typical - check label)	10.0	10.0	6.0	600.0	AC 230 V	Yes	
Ceiling fan (typical - check label)	75.0	4.0	10.0	3000.0	AC 230 V	Yes	Yes
LED TV (32-43 inch) (typical - check label)	60.0	1.0	5.0	300.0	AC 230 V		
Rice cooker (typical - check label)	700.0	1.0	1.0	700.0	AC 230 V		
Water pump 0.5 HP (typical - check label)	450.0	1.0	1.0	450.0	AC 230 V		Yes

Monthly generation vs use

Month	Sun hours	Solar (kWh)	Use (kWh)	Surplus (kWh)
Jan	5.69	311.3	203.1	108.2
Feb	6.28	310.3	183.4	126.9
Mar	6.7	366.5	203.1	163.5
Apr	6.21	328.8	196.5	132.3
May	5.17	282.8	203.1	79.8
Jun	4.1	217.1	196.5	20.6
Jul	3.58	195.9	203.1	-7.2
Aug	3.89	212.8	203.1	9.8
Sep	4.24	224.5	196.5	28.0
Oct	5.28	288.9	203.1	85.8
Nov	5.76	305.0	196.5	108.5
Dec	5.56	304.2	203.1	101.1

Assumptions

- Method and step order from Boxwell, Solar Electricity Handbook (2012); sunshine is site data (PVGIS), not the book's UK/US tables.
- Controller factor 0.9 (MPPT) as in the book (§58).
- Temperature loss 10.9% from the site's PVGIS model (replaces the book's ×1.4 roof rule).
- Battery: LiFePO4 (lithium), depth of discharge 90%, efficiency 95%.
- Book cable formula assumes copper and a 5 % voltage drop; final cables must also meet current-rating rules.
- Grid is available most of the time; the battery only covers the chosen power-cut hours.

Warnings and limitations

- Motor loads (Refrigerator (typical - check label), Ceiling fan (typical - check label), Water pump 0.5 HP (typical - check label)) need a high start-up surge rating - check the inverter's surge (peak) rating against the appliance datasheets (§91).

Data sources and references

- Boxwell, Solar Electricity Handbook (2012)
- PVGIS v5.3 (EC JRC) monthly in-plane irradiation for the site

2. NEA bill and savings (month by month)

Input parameters

NEA meter	15 A
Export credit (net billing)	5.94 Rs/kWh

Calculation method and equations

- Bill = NEA service charge + progressive slab energy charge (ERC 2078 p.74)
- Bill with solar = bill(grid units) − exported units × export credit

Results (calculated by deterministic code)

Estimated yearly saving	21027 Rs/yr
Estimated installed cost (indicative)	271,806.89 Rs

Monthly bills

Month	Use (kWh)	Solar (kWh)	From grid (kWh)	To grid (kWh)	Bill without solar (Rs)	Bill with solar (Rs)	Saving (Rs)
Jan	203	311	120	228	1884	0	1884
Feb	183	310	104	231	1697	0	1697
Mar	203	367	112	276	1884	0	1884
Apr	196	329	106	239	1822	0	1822
May	203	283	109	189	1884	0	1884
Jun	196	217	110	130	1822	223	1599
Jul	203	196	117	110	1884	414	1470
Aug	203	213	117	127	1884	315	1569
Sep	196	224	114	142	1822	194	1628
Oct	203	289	116	202	1884	0	1884
Nov	196	305	115	224	1822	0	1822
Dec	203	304	119	221	1884	0	1884

Assumptions

- Hourly split of solar into self-use and export from an average day per month (PVGIS profile).
- Household load shape: Home - morning and evening peaks.

Data sources and references

- ERC, NEA Electricity Consumer Tariff Determination 2078 (decided 2078/07/08 BS), p.74 [V] - https://www.erc.gov.np/charge-order
- NEPQA-2025 and Amendment-1 (AEPC, Aug 2025 / Apr 2026) [V]
- Net billing export Rs 5.94/kWh - Kathmandu Post 2022; Urja Khabar Jul 2026 [S]

Review record

To be completed by the reviewing engineer. Signing below records review only; it does not by itself constitute certification under any jurisdiction's professional engineering rules.

Reviewed by	
Registration / licence no.	
Signature	
Date	
Comments	